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$$\begin{aligned}f(x) &= \frac{-x^5 + 3x^3 - 2x^2 + x + 1}{x^2} \\&= \frac{-x^5}{x^2} + \frac{3x^3}{x^2} - \frac{2x^2}{x^2} + \frac{x}{x^2} + \frac{1}{x^2} \\&= -x^3 + 3x - 2 + x^{-1} + x^{-2} \\f'(x) &= -3x^2 + 3 - x^{-2} - 2x^{-3} \\&= -3x^2 + 3 - \frac{1}{x^2} - \frac{2}{x^3} \\&= \frac{-3x^5 + 3x^3 - x - 2}{x^3}\end{aligned}$$

References